7 Fig. 7.1 shows some uses of electromagnetic radiation and different regions of the electromagnetic spectrum.

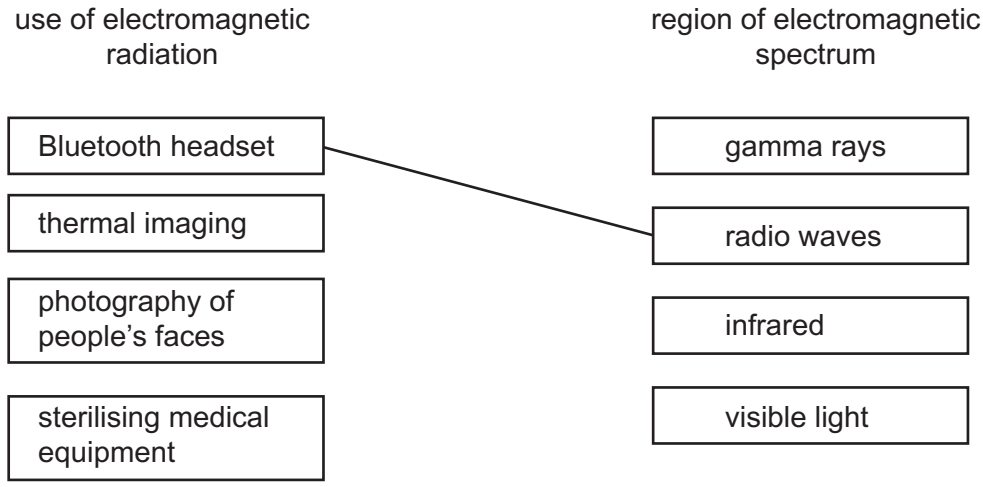


Fig. 7.1

(a) Draw a line from each use to the correct region of the spectrum. Each region of the spectrum is used once. One line has been completed for you. [2]

(b) State the speed of electromagnetic waves in a vacuum.

speed = ... [1]

(c) A Bluetooth headset can be used to listen to music on a mobile (cell) phone without the need for wires to connect the headset to the phone.

(i) The headset uses frequencies in the range 2.40–2.48 GHz.

Calculate the wavelength of the radio waves when the frequency is in the middle of the frequency range.

wavelength = ... [3]

(ii) Suggest why a Bluetooth headset only works well over short distances.

.....

..... [1]

[Total: 7]